# Exit Ticket · reflect and retrieve

## Exit Ticket

True or false: a reference-type argument passed normally allows the method to replace the caller's variable.

## Use key vocabulary from memory

1. Explain by value in your own words.

2. Explain ref in your own words.

3. Explain reference in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Programming Techniques · Value and Reference

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is a local variable?

### Q2

Why use a method?

### Q3

What does an accumulator store?

### Q4

What is a hash collision?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Explain the important decision — An int parameter normally receives a copied number. A Box parameter normally receives a copied reference to the same object. ref instead aliases the caller variable, allowing that variable to be reassigned.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Apply independently

Distinguish passing an object reference by value from using ref. [3]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Apply independently

Explain why changing a normal int parameter does not change the original. [2]

## Question 2

Apply independently

Write a C# method header to increment an integer by reference. [2]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

by value — Copies a value to a parameter

ref — Allows changes to caller variable

reference — Identifies an object

argument — Value provided to a call

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Compare changes to a copy and an object

## Follow the worked method

1. Copy an integer value — Start x=5; CopyInt(x) changes only local n to 9. The caller x remains 5. This is pass-by-value.

2. Copy a reference; mutate one object — Start item→Box A(Value=5), alias=item. Mutate(item) copies the reference, then changes Box A.Value to 9. Both item and alias still refer to Box A.

3. Reassign the caller through ref — Replace(ref item) makes caller item refer to new Box B(Value=20). alias still refers to Box A(Value=9). Without ref, assigning a new Box to local p would leave item unchanged.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Discuss, then respond
If a method takes ref int n and adds 2, starting 3, what is the caller value?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.